BDL Assignment 1: Likelihood and Maximum Likelihood Estimation

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1 Introduction

In case of tossing a coin, $\Omega = \{H,\ T\}$, while $\mathcal{A} = \{\{H\}, \{T\},\ \Omega,\ \varnothing\}$. In case of tossing a coin twice, Ω consists of elementary events $\omega = (\omega_1,\ \omega_2)$, where $\omega_i = H$ or T, i.e., $\Omega = \{(H,H),\ (H,T),\ (T,H),\ (T,T)\}$. And the algebra of events

$$\begin{aligned}\mathcal{A} = \{&\{(H,H)\},\ \{(H,T)\},\ \{(T,H)\},\ \{(T,T)\},\\ &\{(H,H),(H,T)\},\ \{(H,H),(T,H)\},\ \{(H,H),(T,T)\},\\ &\{(H,T),(T,H)\},\ \{(H,T),(T,T)\},\ \{(T,H),(T,T)\},\\ &\{(H,H),(H,T),(T,H)\},\ \{(H,H),(H,T),(T,T)\},\\ &\{(H,H),(T,H),(T,T)\},\ \{(H,T),(T,H),(T,T)\},\\ &\{(H,H),\ (H,T),\ (T,H),\ (T,T)\},\ \varnothing\}.\end{aligned}$$

The event $A = \{(H,H),(H,T)\}$ can be interpreted as "The first flip resulted in heads." The event $B = \{(T,H),(T,T)\}$ can be interpreted as "The first flip resulted in tails." Events A and B are called *mutually exclusive,* because $A \cap B = \varnothing$.

By a **product space** of two probability spaces $(\Omega,\ \mathcal{A},\ P)$, we mean $(\Omega',\ \mathcal{A}',\ P')$, where $\Omega' = \Omega \times \Omega$, i.e., $\omega' = (\omega_1,\ \omega_2)$, where $\omega_1, \omega_2 \in \Omega$. $\mathcal{A}' = \mathcal{A} \otimes \mathcal{A}$ This notation means that we take not simply all Cartesian products $B_1 \times B_2$ with $B_1, B_2 \in \mathcal{A}$, but also complete them to the algebra[1] and P' is defined as

$$P'(\omega' \in B') = P((\omega_1,\omega_2) \in B_1 \times B_2) = P(\omega_1 \in B_1)P(\omega_2 \in B_2).$$

More generally, the N-fold product space of $(\Omega,\ \mathcal{A},\ P)$ with itself models N independent repetitions of the same experiment: $\Omega' = \Omega \times \cdots \times \Omega$ (N times), and the product measure assigns to a product event $B_1 \times \cdots \times B_N$ the probability $P(B_1) \cdots P(B_N)$. This is exactly the structure we use whenever we say that observations $X^{(1)},\ X^{(2)},\ \ldots,\ X^{(N)}$ are *independent and identically distributed (i.i.d.)*: each $X^{(i)}$ lives on its own copy of $(\Omega,\ \mathcal{A},\ P)$, and the joint probability space is the N-fold product space.

Remark 1. *The vector of random variables $X^{(1)},\ X^{(2)},\ \ldots,\ X^{(N)}$ will be denoted by the bold capital letter*

$$\boldsymbol{X} = \big(X^{(1)},\ X^{(2)},\ \ldots,\ X^{(N)}\big),$$

and its realization will be denoted by the lower-case letter

$$\mathbf{x} = \big(\mathrm{x}^{(1)},\ \mathrm{x}^{(2)},\ \ldots,\ \mathrm{x}^{(N)}\big).$$

We will often write $\mathcal{D}$ (from data) for the vector of realizations $\mathbf{x}$.

[1] A collection $\mathcal{A}$ of subsets of Ω is called an **algebra** if it satisfies the following properties:

1. $\Omega \in \mathcal{A}$,
2. if $A \in \mathcal{A}$ and $B \in \mathcal{A}$, then $A \cup B$ and $A \cap B$ belong to $\mathcal{A}$,
3. $A \in \mathcal{A} \ \Rightarrow\ \bar{A} = \Omega \setminus A \in \mathcal{A}$.

The elements of $\mathcal{A}$ are called **events.** The elements $\omega \in \Omega$ are called **elementary events.**

2 Likelihood and log-likelihood

We work with *parametric* statistics: we assume our observations come from a given distribution family $\mathcal{P}_\theta$ with unknown parameter θ taking values in Θ. In the coin example, $\mathcal{P}_\theta = \mathcal{B}er(\theta)$ and $\Theta = [0,\ 1]$.

Definition 1. *Let the joint probability density (or mass) of observations $X^{(1)},\ X^{(2)},\ \ldots,\ X^{(N)}$ depend on parameters $\theta = (\theta_1,\ \theta_2,\ \ldots,\ \theta_K)$:*

$$L(\boldsymbol{x};\ \theta) = p_\theta\big(x^{(1)},\ x^{(2)},\ \ldots,\ x^{(N)}\big). \tag{1}$$

*This is called a **likelihood** function. Viewed as a function of the random variables, $L(\boldsymbol{X};\ \theta)$ is itself a random variable.*

Under the assumption that $X^{(1)},\ \ldots,\ X^{(N)}$ are independent (i.e., that they live on the N-fold product space introduced above),

$$L(\boldsymbol{x};\ \theta) = p_\theta\big(x^{(1)}\big) \cdot p_\theta\big(x^{(2)}\big) \cdot \ldots \cdot p_\theta\big(x^{(N)}\big). \tag{2}$$

Definition 2. *When $p_\theta\big(x^{(i)}\big) > 0$ for all i, one can take the logarithm of the likelihood (2),*

$$\ell(\boldsymbol{x};\ \theta) = \sum_{i=1}^{N} \log p_\theta\big(x^{(i)}\big), \tag{3}$$

*which is called the **log-likelihood** function.*

The *Principle of Maximum Likelihood* states: if we regard the likelihood function, evaluated at the observed realizations of RVs $X^{(1)},\ \ldots,\ X^{(N)}$, as a function of the parameter θ, then the best estimate of θ is the maximizer of this function:

$$\theta^*_{ML} = \underset{\theta\in\Theta}{\operatorname{argmax}}\ L(\mathcal{D};\ \theta). \tag{4}$$

Because the logarithm is monotonically increasing, this is equivalent to

$$\theta^*_{ML} = \underset{\theta\in\Theta}{\operatorname{argmax}}\ \ell(\mathcal{D};\ \theta). \tag{5}$$

Worked example: the coin

Assuming independence of $X^{(1)},\ \ldots,\ X^{(N)} \sim \mathcal{B}er(\theta)$, the joint distribution is

$$p(\boldsymbol{x}) = p\big(x^{(1)},\ x^{(2)},\ \ldots,\ x^{(N)}\big) = \prod_{i=1}^{N} \theta^{x^{(i)}}(1-\theta)^{1-x^{(i)}} = \theta^{\sum_{i=1}^{N} x^{(i)}}(1-\theta)^{N-\sum_{i=1}^{N} x^{(i)}}. \tag{6}$$

If $x^{(i)} = 1$ (heads) occurs m times and $x^{(i)} = 0$ (tails) occurs $N-m$ times, then

$$L(\mathcal{D};\ \theta) = \theta^{\sum_{i=1}^{N} \mathrm{x}^{(i)}}(1-\theta)^{N-\sum_{i=1}^{N} \mathrm{x}^{(i)}} = \theta^m(1-\theta)^{N-m}, \tag{7}$$

and the corresponding log-likelihood is

$$\ell(\boldsymbol{x};\ \theta) = \sum_{i=1}^{N} x^{(i)} \log\theta + \left(N - \sum_{i=1}^{N} x^{(i)}\right) \log(1-\theta) = m\log\theta + (N-m)\log(1-\theta). \tag{8}$$

Task 1

Consider N independent tosses of the same coin, modeled as i.i.d. random variables $X^{(1)},\ \ldots,\ X^{(N)} \sim \mathcal{B}er(\theta)$, of which m come up heads. Using the Principle of Maximum Likelihood, determine the estimate θ^*_{ML} of the parameter θ.

3 Conditional probability, marginals, and the chain rule

Definition 3 (Conditional probability)**.** *For events $A,\ B$ with $P(B) > 0$, the conditional probability of A given B is*

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

For random variables $X,\ Y$ with joint density $p(x, y)$, the conditional density of X given $Y = y$ (whenever $p(y) > 0$) is

$$p(x \mid y) = \frac{p(x, y)}{p(y)}. \tag{9}$$

Definition 4 (Marginal distribution)**.** *Given a joint density $p(x, y)$, the **marginal** density of Y is obtained by integrating (or, in the discrete case, summing) X out:*

$$p(y) = \int p(x, y)\, \mathrm{d}x. \tag{10}$$

Proposition 1 (Chain rule)**.** *For any random variables (or vectors) $X,\ Y$, rearranging* (9) *gives*

$$p(x, y) = p(x \mid y)\, p(y) = p(y \mid x)\, p(x). \tag{11}$$

More generally, for $X^{(1)},\ \ldots,\ X^{(N)}$,

$$p\big(x^{(1)},\ \ldots,\ x^{(N)}\big) = p\big(x^{(1)}\big)\, p\big(x^{(2)} \mid x^{(1)}\big)\, p\big(x^{(3)} \mid x^{(1)}, x^{(2)}\big) \cdots p\big(x^{(N)} \mid x^{(1)},\ \ldots,\ x^{(N-1)}\big).$$

Note that when $X^{(1)},\ \ldots,\ X^{(N)}$ are independent, every conditional factor collapses to a marginal, $p\big(x^{(i)} \mid x^{(1)},\ \ldots,\ x^{(i-1)}\big) = p\big(x^{(i)}\big)$, and the chain rule reduces to the product in (2)*.*

Now treat θ itself as random, with joint density $p(\mathcal{D},\ \theta)$ between the data $\mathcal{D}$ and the parameter θ. Using the definition of conditional probability (9), the chain rule (11), and the marginal (10), derive **Bayes' rule**:

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)\, p(\theta)}{p(\mathcal{D})}.$$

We can write $p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta)\, p(\theta)\, \mathrm{d}\theta$ explicitly, and identify $p(\mathcal{D},\ \theta)$ as the likelihood $L(\mathcal{D};\ \theta)$ from Section 2, the **prior** $p(\theta)$, the **posterior** $p(\theta \mid \mathcal{D})$, and the **marginal likelihood** (or **evidence**) $p(\mathcal{D})$.

Task 2

Now suppose θ itself is treated as random, with a prior distribution $\theta \sim \mathrm{Beta}(a,\ a)$ for a known $a > 0$, i.e.,

$$p(\theta) = \frac{1}{B(a, a)}\, \theta^{a-1}(1 - \theta)^{a-1}, \qquad \theta \in [0, 1].$$

Given the same data $\mathcal{D}$ (m heads out of N i.i.d. tosses), determine the parameter θ by finding the estimate that maximizes the *posterior* density $p(\theta \mid \mathcal{D}) \propto L(\mathcal{D};\ \theta)\, p(\theta)$ (i.e., the maximum a posteriori, or MAP, estimate θ^*_{MAP}).

Note. Write out $p(\theta \mid \mathcal{D})$ up to normalization, take its logarithm, and solve for the maximizer.

Compare θ^*_{MAP} to θ^*_{ML} from Task 1, and discuss what happens to θ^*_{MAP} as $a \to 1$ and as $N \to \infty$ with m/N fixed.